

ENTERPRISE LEGAL TECHNOLOGY ARCHITECTURE

LEX REGIS

Enterprise-Grade AI-Powered Legal Practice Management, Judicial Intelligence & Citizen Access Platform

SYSTEM ARCHITECTURE

Modular Monolith & Layered Domain

ARTIFICIAL INTELLIGENCE

Groq Cloud LLM (openai/gpt-oss-120b)

INTEGRITY & CRYPTOGRAPHY

SHA-256 Checksums / Blockchain Receipts

CORE TECHNOLOGY STACK

Python 3.13 / Django 5.x / Daphne ASGI

REAL-TIME SUBSYSTEM

Django Channels / Redis WebSockets

JURISDICTION ENGINE

Indian Legal Framework & Global RBAC

Document Classification: Complete Software Engineering Project Report (Ready for Submission)

Date: August 2026 | **Release Version:** 1.0.0-PROD-READY | **Status:** Validated & Approved

LEX REGIS: Enterprise-Grade AI-Powered Legal Technology Platform

Comprehensive Technical Project Report & System Architecture Specification

Project Metadata

- **Project Title:** LEX REGIS – Enterprise Legal Practice Management, Judicial Intelligence & Citizen Access System
 - **Domain:** Legal Technology (LegalTech), Artificial Intelligence, Machine Learning & Cryptographic Distributed Ledger Systems
 - **Architecture:** Modular Monolithic Architecture with Layered Service-Selector Pattern & Asynchronous Event Engine
 - **Technology Stack:** Python 3.13+, Django 5.x, Django REST Framework, Django Channels (ASGI/Daphne), Redis, Groq Cloud LLM API, SQLite / PostgreSQL, HTMX, Alpine.js
 - **Document Classification:** Comprehensive Software Engineering Project Report & System Documentation (Ready for Academic / Industrial Submission)
 - **Date of Submission:** August 2026
 - **Version:** 1.0.0-PROD-READY
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Executive Summary

The modern legal ecosystem is characterized by overwhelming procedural friction, opaque case tracking, severe judicial backlog, fragmented communication between litigants and advocates, and an absence of verifiable evidentiary integrity. **LEX REGIS** is an enterprise-grade, full-stack LegalTech platform engineered to bridge the critical gap between citizens seeking justice and legal professionals delivering counsel.

LEX REGIS combines an intuitive citizen intake wizard, an AI-powered legal triage engine, an advocate practice management suite, a court hearing scheduler with conflict detection, a cryptographic document vault with simulated blockchain immutability, and real-time WebSocket communication channels. Built on a modular monolithic Django architecture adhering strictly to the **Service-Selector Pattern**, **BaseModel Mixin inheritance**, and **Role-Based Access Control (RBAC)**, the system maintains strict separation of concerns, enterprise scalability, and compliance with jurisdictional legal standards (including the Indian Legal Framework).

This comprehensive document serves as the formal project report and complete technical specification, detailing the system requirements, architectural diagrams, domain models, algorithms, security frameworks, API contracts, testing methodologies, and deployment blueprints.

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Chapter 1: Problem Statement, Background & Project Objectives

1.1 Background & Context

Across global jurisdictions, particularly in rapidly digitizing legal systems such as India, the legal sector suffers from structural inefficiencies:

- **Judicial Case Overload:** Millions of cases remain pending across District Courts, High Courts, and the Supreme Court. A significant portion of delays is caused by manual scheduling errors, repeated adjournments, and inefficient docket management.
- **Asymmetry of Information for Citizens:** Litigants often cannot evaluate the merits, timeline, or cost of their legal matters prior to hiring counsel, leading to exploitation, mismatched legal representation, and unnecessary litigation.
- **Fragmented Practice Tools for Advocates:** Lawyers and law firms juggle disconnected software tools for drafting, billing, hearing calendars, client communication, and document storage.
- **Evidentiary Vulnerability:** Legal documents, affidavits, and case timeline logs are susceptible to tampering, loss, and unauthorized alterations in traditional file-storage environments.

1.2 Problem Statement

"To design, architect, and implement an end-to-end, multi-tenant enterprise Legal Technology platform that unifies citizen legal intake, AI-driven preliminary case analysis, intelligent lawyer matchmaking, comprehensive case lifecycle management, automated hearing management, tamper-evident document storage via cryptographic hashing and blockchain verification, and real-time client-lawyer collaboration into a single, highly secure, modular web ecosystem."

1.3 Project Objectives

1. **Intelligent Citizen Intake:** Develop a guided multi-step wizard allowing citizens to articulate legal grievances, submit documents, and receive real-time AI-assisted statutory categorizations, risk scores, and estimated budgets.
2. **Automated Legal Analysis & Lawyer Matchmaking:** Leverage cutting-edge Large Language Models (Groq API / LLaMA / DeepSeek models) to extract applicable Acts, Sections, keywords, and match matters with verified advocates based on practice domain and geographic jurisdiction.
3. **Enterprise Case Workspace:** Implement a comprehensive 13-stage state-machine case tracking engine incorporating financial retainers, multi-party details, task delegation, timeline audit logs, and courtroom assignments.
4. **Hearing Docket Management:** Create a court hearing and calendar engine with automated adjournment tracking, delay metrics for ML training, and participant conflict detection.
5. **Cryptographic Proof of Authenticity:** Establish a document management vault featuring SHA-256 fingerprinting, versioning, automated metadata extraction, and verifiable ledger logging (`BlockchainRecord`) to establish non-repudiation.
6. **Real-Time Synchronous Communication:** Implement WebSocket-based bidirectional chat rooms and real-time push alerts powered by Django Channels, Daphne, and Redis.
7. **Judicial & Firm Analytics:** Provide high-resolution analytical dashboards visualizing national disposal rates, lawyer performance metrics, and court congestion indices.

Chapter 2: Literature Review & Comparative Market Analysis

2.1 Review of Existing Systems

Traditional Legal Practice Management (LPM) and Legal Aid solutions exhibit distinct functional gaps:

FEATURE / DIMENSION	TRADITIONAL LPM (E.G. CLIO / PRACTICEPANTHER)	E-COURTS PORTAL (GOVT OF INDIA)	GENERIC ERP SYSTEMS	
Target Audience	Law Firms / Solo Lawyers only	Courts & Litigants (Read-only)	General Enterprises	Unified: Citizens, Advocates, Law Firms, Admins
Intake AI Triage	✗ No automated AI statutory analysis	✗ None	✗ None	✓ Groq LLM Zero-Shot Legal Triage & Act Extraction
Lawyer Matching	✗ None (Internal firm cases only)	✗ None	✗ None	✓ Dynamic AI Practice-Area & Proximity Matching
Case State Engine	⚠ Basic task statuses	⚠ Fixed procedural stages	⚠ Generic workflow	✓ 13-Stage Legal Lifecycle with State Validation
Document Integrity	⚠ Basic S3 / Cloud Storage	⚠ Unsigned PDF attachments	⚠ Standard blob storage	✓ SHA-256 Checksums + Blockchain Receipt Ledger
Real-time Comms	⚠ Static email notifications	✗ SMS only (One-way)	⚠ Webhook notifications	✓ Live WebSocket Messaging + In-App Push Engine
Jurisdiction Engine	✗ Western / US Law focused	✓ Indian Law specific	✗ Generic	✓ Built-in Indian & Global Statutory Validators (Aadhaar, PAN, Bar Reg, CNR)
Predictive ML	✗ None	✗ None	✗ None	✓ Duration & Risk ML Model Infrastructure

2.2 Feasibility Study

- **Technical Feasibility:** Python 3.13 and Django 5.x offer an optimal combination of ORM maturity, security guarantees, and high-performance asynchronous execution via ASGI (Daphne). Groq Cloud API delivers ultra-low-latency (<1.5s) inference for complex legal analysis.
- **Economic Feasibility:** The architecture utilizes open-source components (Django, SQLite/PostgreSQL, Redis, HTMX, Tailwind/Bootstrap) and cost-effective serverless LLM endpoints, eliminating vendor lock-in.
- **Operational Feasibility:** Role-specific interfaces ensure minimal learning curves for non-technical citizens while providing high-density workspaces for legal professionals.

Chapter 3: System Requirements Specification (SRS)

3.1 User Roles & Personas

The system strictly distinguishes four primary personas governed by `RoleChoices`:

1. **Client / Citizen (`CLIENT`)**: Submits grievances via the AI Intake Wizard, reviews AI legal summaries, books consultations with verified lawyers, tracks case timelines, uploads evidence, and communicates in real time.
2. **Advocate (`ADVOCATE`)**: Manages assigned matters, conducts case intake evaluations from the marketplace, schedules hearings, tracks billable hours, uploads legal pleadings, and generates invoices.
3. **Law Firm (`LAW_FIRM`)**: Oversees multiple associate advocates, assigns incoming matters to partners, monitors organizational revenue, and manages institutional corporate clients.
4. **System Administrator (`ADMIN`)**: Manages user verification (Bar Council verification, ID validation), oversees system security logs, configures master data (Courts, Acts, Sections), and audits AI request telemetry.

```
graph TD
    User([User Registration]) --> RoleDecision{Role Selection}
    RoleDecision -->|Citizen| ClientPersona[Client / Citizen Workspace]
    RoleDecision -->|Advocate| LawyerPersona[Lawyer Portal & Marketplace]
    RoleDecision -->|Law Firm| FirmPersona[Law Firm Enterprise Dashboard]
    RoleDecision -->|Administrator| AdminPersona[Platform Administration & Audit]

    ClientPersona --> Intake[AI Intake Wizard]
    Intake --> AIA[AI Analysis Groq Legal Analysis Engine]
    AIA --> LawyerMatching[AI Advocate Recommendations]
    LawyerMatching --> CaseCreation[Enterprise Case Lifecycle]

    LawyerPersona --> CaseAccept[Case Acceptance & Assignment]
    CaseAccept --> CaseCreation

    CaseCreation --> DocVault[Cryptographic Document Vault]
    CaseCreation --> HearingScheduler[Hearing & Adjournment Engine]
    CaseCreation --> LiveComms[WebSocket Chat Channels]
    DocVault --> BlockchainLedger[Blockchain Immutability Engine]
```

3.2 Functional Requirements (FR)

- **FR-01 (Authentication & Profile Management)**: Multi-step user onboarding, email verification, role-specific profile generation (`Profile` , `ProfessionalProfile` , `Organization`), and granular security settings (2FA, session expiry).
- **FR-02 (AI Legal Intake Wizard)**: Multi-step grievance capture with dynamic file upload, automatic language detection, and background transmission to Groq AI.
- **FR-03 (Statutory Analysis & Scoring)**: Real-time JSON extraction of legal domain, complexity, suggested documents, applicable Acts/Sections, estimated budget, timeline, and risk level.
- **FR-04 (Intelligent Advocate Matchmaking)**: Semantic matching algorithm ranking lawyers based on domain expertise, jurisdiction, experience, and fee budget.
- **FR-05 (Enterprise Case Workspace)**: Complete case docketing with unique alphanumeric case numbers (`LR-YYYY-XXXXXX`), financial records, opposite party tracking, and audit trails.

- **FR-06 (State Transition Management):** Strict 13-stage lifecycle enforcement (`DRAFT` → `AI_ANALYSED` → `PENDING_ACCEPTANCE` → `LAWYER_ASSIGNED` → `CONSULTATION` → `EVIDENCE` → `LEGAL_NOTICE` → `PETITION` → `FILED` → `HEARINGS` → `JUDGEMENT` → `CLOSED` → `ARCHIVED`).
- **FR-07 (Document Integrity Vault):** File upload handler computing SHA-256 digests, extracting MIME types, enforcing size quotas, and maintaining immutable version chains (`DocumentVersion`).
- **FR-08 (Cryptographic Verification Ledger):** Registration of document and event hashes into `BlockchainRecord` with simulated Ethereum transaction receipts and IPFS CID storage.
- **FR-09 (Court Hearing & Adjournment Engine):** Hearing scheduling with court room assignment, judge availability checking, outcome recording, and adjournment reason logging.
- **FR-10 (Real-Time Communication):** Authenticated ASGI WebSocket channels for instant client-advocate messaging with typing status and persistence.
- **FR-11 (Multi-Channel Notifications):** System-wide event broadcasting for case updates, hearing reminders, consultation requests, and document approvals.
- **FR-12 (Judicial & Firm Analytics):** Real-time KPI aggregation for case health, disposal velocity, court congestion indices, and lawyer billing throughput.

3.3 Non-Functional Requirements (NFR)

- **NFR-01 (Performance & Latency):** Web page load times under 800ms; AI intake analysis completed within 2.5 seconds; WebSocket round-trip message delivery under 50ms.
- **NFR-02 (Security & Confidentiality):** Attorney-client privilege safeguarded via object-level permissions, encrypted credentials, CSRF tokens on all modifying requests, parameterized ORM queries, and SHA-256 password hashing (PBKDF2/Argon2).
- **NFR-03 (Reliability & Data Integrity):** Zero data loss via database-level transaction atomicity (`@transaction.atomic`), soft-delete mixin architecture, and automated daily backup scripts.
- **NFR-04 (Scalability):** Stateless application tier capable of horizontal scaling behind Nginx load balancers, with Redis-backed session and WebSocket channel layers.
- **NFR-05 (Maintainability):** Adherence to PEP8, strong Python 3 type hinting (`typing`), modular app separation, and the Service-Selector architecture.

Chapter 4: System Architecture & Design Philosophy

4.1 Modular Monolithic Architecture

LEX REGIS adopts a **Modular Monolith** pattern. Rather than introducing the network latency and distributed failure modes of microservices prematurely, domain logic is partitioned into self-contained Django applications. Each app encapsulates its own domain models, data selectors, business services, presentation layers, forms, and API serializers.

```
c:\Users\vinay\Desktop\Project\lex_regis\
├── apps/
│   ├── common/           # Core architectural base models, mixins, validators, exceptions
│   ├── accounts/         # Authentication, RBAC, User & Profile domain
│   ├── intake/           # Citizen intake wizard & AI consultation requests
│   ├── cases/            # Core Case domain, State Machine, Financials, Parties
│   ├── documents/        # Cryptographic Document Vault, Checksums, Versions
│   ├── hearings/         # Courtroom docketing, schedules, adjournments
│   ├── ai/               # Groq LLM integration, prompts, audit logs, AI chat
│   ├── ml/               # Predictive models (duration, success, court workload)
│   ├── blockchain/       # Tamper-evident ledger records & verification
│   ├── communication/    # Django Channels ASGI WebSockets, live chat rooms
│   ├── notifications/    # Notification dispatch & in-app alerts
│   ├── dashboard/        # Role-specific analytics, KPI cards, global search
│   ├── analytics/        # Judicial statistics & national case metrics
│   ├── lawyer_portal/    # Advocate workspace, 9-step registration wizard
│   └── public_site/      # Public landing pages, lawyer directory, marketing
├── config/               # Environment-aware settings, ASGI/WSGI, root routing
├── templates/            # Global & app-specific semantic HTML templates
├── static/               # Static assets (CSS, JavaScript, images)
├── media/                # Encrypted user uploads & evidence documents
└── tests/                # Comprehensive test suites (unit, integration, API)
```

4.2 The Service-Selector Layered Architecture

To prevent "fat models" and "fat views", LEX REGIS enforces strict architectural layering:

```
graph TD
    ClientReq[HTTP Request / WebSocket Frame] --> PresentationLayer[Presentation Layer: Views / Consum
    PresentationLayer --> SecurityLayer[Permission Classes / Role Guards]
    SecurityLayer --> DecisionFlow[Operation Type]

    DecisionFlow -->|Read Query| SelectorLayer[Selectors Layer: pure functions returning QuerySets / D
    DecisionFlow -->|Write / Mutation| ServiceLayer[Services Layer: Business Logic, Atomic Transaction

    SelectorLayer --> ORM[Django ORM Layer]
    ServiceLayer --> ORM
    ServiceLayer --> ExternalServices[External Services: Groq LLM, Redis, Blockchain]

    ORM --> Database[(SQLite / PostgreSQL Database)]
```

1. **Presentation Layer** (`presentation/views.py` , `api/views.py`): Responsible only for parsing HTTP requests, validating form/serializer input, and returning HTTP/JSON responses.
2. **Service Layer** (`services/`): Contains all domain mutation logic, external API calls (e.g., Groq Cloud), transaction boundaries (`@transaction.atomic`), and business rules.
3. **Selector Layer** (`selectors/`): Contains optimized, reusable database read queries, applying filtering, prefetching (`select_related` , `prefetch_related`), and complex aggregations.
4. **Model Layer** (`models/`): Encapsulates relational schema definitions, constraints, database indexes, and model property methods.

4.3 BaseModel & Mixin Inheritance Hierarchy

All core domain models inherit from `apps.common.models.base.BaseModel` , ensuring consistent enterprise behaviors across the entire platform:

```

classDiagram
    class UUIDMixin {
        +UUID id
    }
    class TimestampMixin {
        +DateTime created_at
        +DateTime updated_at
    }
    class SoftDeleteMixin {
        +Boolean is_deleted
        +DateTime deleted_at
        +User deleted_by
        +soft_delete(user)
        +restore()
    }
    class OwnershipMixin {
        +User created_by
        +User updated_by
    }
    class AuditMixin {
        +Boolean is_active
    }
    class BaseModel {
        +objects Manager
        +active_objects ActiveManager
        +available_objects SoftDeleteManager
        +hard_delete()
        +restore()
        +is_recently_updated()
    }

    UUIDMixin <|-- BaseModel
    TimestampMixin <|-- BaseModel
    SoftDeleteMixin <|-- BaseModel
    OwnershipMixin <|-- BaseModel
    AuditMixin <|-- BaseModel

    BaseModel <|-- User
    BaseModel <|-- Case
    BaseModel <|-- Document
    BaseModel <|-- Hearing
    BaseModel <|-- LegalIssue
    BaseModel <|-- BlockchainRecord
    BaseModel <|-- CasePrediction

```

Chapter 5: Detailed App-by-App Modular Subsystem Decomposition

5.1 apps.accounts : Identity, RBAC & Profile Management

The accounts subsystem manages user identity, role-specific attributes, contact addresses, organizational hierarchies, and security parameters.

- **Custom User Model (User)**: Subclasses `AbstractBaseUser` and `PermissionsMixin`. Uses `email` as the primary unique username field. Stores system attributes such as `user_code`, `failed_login_attempts`, `login_count`, and `account_status`.
- **User Roles (RoleChoices)**: `ADMIN`, `ADVOCATE`, `LAW_FIRM`, `CLIENT`.
- **Profiles Ecosystem**:
 - `Profile`: General profile storing client type (Individual/Corporate), organization name, Aadhaar/PAN details, and communication preferences.
 - `ProfessionalProfile`: Lawyer-specific profile storing Bar Council Registration Number

(`XX/12345/YYYY`), designation, law firm affiliation, years of experience, and practice areas.

- **Organization** : Corporate entity/law firm profile with registration number, tax identifier, and employee rosters.
- **SecuritySettings** : Tracks two-factor authentication, login session limits, and password expiry policies.

5.2 `apps.intake` : Citizen AI Grievance Wizard & Lawyer Matching

The intake module empowers citizens to navigate legal complexity without initial technical jargon:

- **LegalIssue Model**: Captures citizen narrative, incident date, state, district, matter value, urgency (`ROUTINE` , `URGENT` , `EMERGENCY`), and current analysis status.
- **DocumentUpload Model**: Handles supporting grievance documents (contracts, FIRs, notices).
- **AIAnalysis Model**: Persists the structured evaluation produced by Groq AI, including statutory category, practice area, complexity, suggested documents, applicable Acts, applicable Sections, next steps, cost estimates, and risk level.
- **Recommendation Model & Matchmaker**: Computes match scores between legal issues and verified `ProfessionalProfile` records, presenting top advocate matches.
- **ConsultationRequest Model**: Manages formal booking requests from citizens to advocates, capturing snapshot pricing and initiating case creation upon lawyer acceptance.

5.3 `apps.cases` : Enterprise Case Management & State Engine

The core domain of LEX REGIS, handling the complete lifecycle of legal matters:

- **Case Model**: Features unique human-readable case numbers (`CASE-XXXXXXXX`), full case narrative, AI summaries, client references, assigned advocate/firm links, practice classification, case valuation, and court venue.

- 13-Stage State Machine:

```
text [DRAFT] → [AI_ANALYSED] → [PENDING_ACCEPTANCE] → [LAWYER_ASSIGNED] | [PETITION] ← [LEGAL_NOTICE] ← [EVIDENCE] ← [CONSULTATION] | ↳ [FILED] → [HEARINGS] → [JUDGEMENT] → [CLOSED] → [ARCHIVED]
```

- **CaseParty Model**: Tracks multiple petitioners, respondents, third parties, and opposing legal counsels.
- **CaseFinancial Model**: Manages retainer agreements, consultation fees, court fees, stamp duties, GST calculation, discounts, and payments.
- **CaseTimeline Model**: Immutable event stream recording every lifecycle event, status change, document addition, or hearing result with actor attribution.
- **CaseTask Model**: Task assignment and delegation with deadlines, priority levels, and completion status.

5.4 `apps.documents` : Cryptographic Document Management Vault

The document vault ensures file integrity, categorization, and auditability:

- **Document Model**: Links uploaded files to specific cases, owners, and document types.
- **SHA-256 Checksum Calculation**: Files are processed through `hashlib.sha256` upon upload to create a unique cryptographic fingerprint, preventing file corruption or silent tampering.
- **DocumentVersion Model**: Maintains complete historical snapshots whenever a document is modified or updated.
- **DocumentMetadata Model**: Stores extracted text, OCR results, and page metrics.
- **Master Classifications**: `DocumentType` , `DocumentCategory` , `DocumentStatus` , `DocumentVisibility` .

5.5 `apps.hearings` : Court Docketing & Adjournment Management

Controls court proceedings, calendar schedules, and hearing analytics:

- **Hearing Model:** Docket records with scheduled date, time, court room, presiding judge, estimated duration, actual start/end times, and outcome summary.
- **Delay & ML Feature Tracking:** Automatically calculates `delay_minutes` (discrepancy between scheduled and actual start time) and `adjournment_count` for predictive machine learning models.
- **Adjournment Model:** Formal logs of hearing postponements, capturing requesting party, justification code, and next scheduled date.
- **JudgeSchedule Model:** Courtroom availability and holiday management (`CourtHoliday`).

5.6 `apps.ai` : Groq LLM Intelligence & Diagnostic Telemetry

Integrates high-speed inference for legal analysis and conversational support:

- **Groq Cloud API Client:** Configured with `openai/gpt-oss-120b`, temperature 0.1 for deterministic statutory extraction, and temperature 0.3 for conversational legal assistance.
- **Conversation & Message Models:** Multi-turn legal chat history linked optionally to a specific `Case` or `Document`.
- **AIRequestLog Model:** Comprehensive observability table recording every API call, prompt tokens, completion tokens, total tokens, latency in milliseconds (`latency_ms`), status codes, response payloads, and exception tracebacks.

5.7 `apps.ml` : Predictive Analytics & Workload Modeling

Machine learning feature stores and statistical prediction representations:

- **CasePrediction Model:** Stores machine-generated predictions for case duration in days (`duration_days_predicted`), success probability percentage (`success_probability`), risk score (`risk_score`), and delay likelihood (`delay_probability`).
- **CourtWorkload Model:** Computes court congestion indices (`congestion_index`), active case counts, and average disposal turnaround times.

5.8 `apps.blockchain` : Distributed Ledger & Verification Subsystem

Provides cryptographic proof of authenticity for sensitive legal records:

- **BlockchainRecord Model:** Utilizes Django's `GenericForeignKey` framework (`ContentType`, `object_id`) to bind to any entity (Document, Case Event, Evidence, Hearing).
- **Data Payload Serialization:** Generates sorted JSON canonical representations of the target object, computes its SHA-256 fingerprint, generates simulated Ethereum transaction hashes, and records verification timestamps.

5.9 `apps.communication` : Live WebSocket Messaging Engine

Facilitates real-time, zero-latency interactions between clients and lawyers:

- **Daphne & ASGI Pipeline:** Handles bidirectional WebSocket connections protocol-upgraded from HTTP.
- **ChatConsumer (AsyncWebsocketConsumer):** Validates user authentication, verifies room participation permissions asynchronously (`database_sync_to_async`), broadcasts messages to Redis channel groups (`channel_layer.group_send`), and commits chat logs to the database.

5.10 `apps.notifications` : Multi-Channel System Alerts

- **Notification Model:** Manages user alerts categorized by type (`CASE_UPDATE` , `HEARING_REMINDER` , `DOCUMENT_UPLOAD` , `LAWYER_ASSIGNMENT` , `PAYMENT` , `SYSTEM_ALERT`).
- **In-App & Email Dispatch:** Provides notification badges, unread counts, and direct action links.

5.11 `apps.dashboard` & `apps.analytics` : Business Intelligence & Judicial Metrics

- **DashboardPresenter** : Compiles role-aware KPI cards, upcoming hearings, recent activity feeds, and quick action bars.
- **AnalyticsService** : Aggregates national legal statistics, state-by-state case distributions, disposal growth rates, and category breakdowns.

5.12 `apps.lawyer_portal` : Advocate Workspace & 9-Step Registration

- **Lawyer Workspace:** Centralized management of accepted cases, pending intake consultations, calendar dockets, and billing summaries.
- **9-Step Registration Wizard:** Comprehensive advocate onboarding capturing personal bio, Bar Council credentials, educational degrees (`LawyerEducation`), practice areas (`LawyerPracticeArea`), office locations (`LawyerOffice`), fee structures (`LawyerFees`), availability (`LawyerAvailability`), and verification documents (`LawyerVerification`).

5.13 `apps.common` : Core Infrastructure, Utilities & Legal Validators

- **Indian Legal Validators (`indian_legal.py`):**
 - Aadhaar Validator: 12-digit regex validation (`^\d{12}$`).
 - PAN Card Validator: Alphanumeric format (`^[A-Z]{5}[0-9]{4}[A-Z]{1}$`).
 - Bar Council Registration: State format (`^[A-Z]{2}/\d{1,5}/\d{4}$`).
 - Case Number Validator: Alphanumeric standard (`^LR-\d{4}-\d{6}$`).
 - **Error Handlers:** Custom presentation templates for HTTP 403 (Forbidden), 404 (Not Found), and 500 (Internal Server Error).
-

Chapter 6: Comprehensive Database Design & Entity Relationship Architecture

6.1 Relational Entity Relationship Diagram (ERD)

```

erDiagram
    USER ||--o{ PROFILE : has
    USER ||--o{ PROFESSIONAL_PROFILE : possesses
    USER ||--o{ ORGANIZATION : manages
    USER ||--o{ NOTIFICATION : receives
    USER ||--o{ AI_REQUEST_LOG : triggers

    PROFILE ||--o{ CASE : owns_as_client
    PROFESSIONAL_PROFILE ||--o{ CASE : assigned_as_lawyer
    ORGANIZATION ||--o{ CASE : firm_supervision

    USER ||--o{ LEGAL_ISSUE : submits
    LEGAL_ISSUE ||--o{ AI_ANALYSIS : evaluated_by
    LEGAL_ISSUE ||--o{ DOCUMENT_UPLOAD : attaches
    LEGAL_ISSUE ||--o{ RECOMMENDATION : matches
    LEGAL_ISSUE ||--o{ CONSULTATION_REQUEST : generates

    CASE ||--o{ CASE_PARTY : includes
    CASE ||--o{ CASE_FINANCIAL : records
    CASE ||--o{ CASE_TIMELINE : audits
    CASE ||--o{ CASE_TASK : delegates
    CASE ||--o{ DOCUMENT : contains
    CASE ||--o{ HEARING : schedules
    CASE ||--o{ CASE_PREDICTION : forecasts
    CASE ||--o{ CONVERSATION : establishes

    DOCUMENT ||--o{ DOCUMENT_VERSION : tracks
    DOCUMENT ||--o{ DOCUMENT_METADATA : analyzes
    DOCUMENT ||--o{ BLOCKCHAIN_RECORD : notarizes

    HEARING ||--o{ ADJOURNMENT : logs
    HEARING }|--|| COURT_ROOM : hosted_in

    CONVERSATION ||--o{ MESSAGE : contains
  
```

6.2 Data Dictionaries for Core Models

Table 1: **accounts_user**

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	Primary Key, default=uuid4	Unique user identifier
email	EmailField	Unique, db_index=True	Primary login credential
password	CharField(128)	Not Null	Cryptographic hash of password
first_name	CharField(100)	Not Null	User given name
last_name	CharField(100)	Not Null	User family name
phone_number	CharField(20)	Nullable, Validated	Contact phone number
role	CharField(20)	Choices: ADMIN , ADVOCATE , LAW_FIRM , CLIENT	Primary authorization role
email_verified	Boolean	Default=False	Email confirmation flag
is_active	Boolean	Default=True	Soft activation flag
created_at	DateTime	Auto Now Add, db_index=True	Creation timestamp

Table 2: `cases_case`

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	Primary Key, default=uuid4	Unique case identifier
<code>case_number</code>	CharField(50)	Unique, db_index=True	Alphanumeric case identifier
<code>title</code>	CharField(255)	Not Null	Brief title of the matter
<code>description</code>	TextField	Blank allowed	Complete case narrative
<code>ai_summary</code>	TextField	Blank allowed	AI generated synopsis
<code>client_id</code>	UUID (FK)	References <code>accounts_profile</code> , on_delete=PROTECT	Matter client profile
<code>assigned_lawyer_id</code>	UUID (FK)	References <code>accounts_professionalprofile</code> , SET_NULL	Assigned advocate
<code>matter_category</code>	CharField(100)	Blank allowed	Civil, Criminal, Corporate, etc.
<code>status</code>	CharField(50)	Choices: <code>CaseStatus</code> (13 stages), default= <code>DRAFT</code>	Current lifecycle state
<code>matter_source</code>	CharField(20)	Choices: <code>AI_INTAKE</code> , <code>MANUAL</code>	Matter origin
<code>budget_estimate</code>	Decimal(14,2)	Nullable	Estimated total legal cost
<code>case_value</code>	Decimal(14,2)	Nullable	Financial dispute value
<code>is_deleted</code>	Boolean	Default=False, db_index=True	Soft-delete status

Table 3: documents_document

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	Primary Key, default=uuid4	Unique document identifier
document_number	CharField(100)	Unique, db_index=True	Systematic document tracking code
case_id	UUID (FK)	References cases_case , CASCADE	Parent case docket
uploaded_by_id	UUID (FK)	References accounts_user , SET_NULL	Uploader user account
original_file	FileField	Dynamic upload path	Stored file binary
file_size	PositiveInteger	Default=0	Binary size in bytes
checksum	CharField(64)	SHA-256 Hash	Cryptographic integrity hash
blockchain_verification_status	CharField(50)	Choices: PENDING , VERIFIED , REJECTED	Verification state
blockchain_transaction_hash	CharField(100)	Blank allowed	Ledger transaction identifier

Table 4: ai_airequestlog

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	Primary Key	Telemetry record ID
user_id	UUID (FK)	References accounts_user , SET_NULL	Requesting user
endpoint	CharField(100)	Not Null	e.g. intake.analysis , chat.completions
model_name	CharField(100)	Not Null	e.g. openai/gpt-oss-120b
prompt_tokens	PositiveInteger	Default=0	Input token count
completion_tokens	PositiveInteger	Default=0	Output token count
latency_ms	FloatField	Default=0.0	Request round-trip time in ms
is_success	Boolean	Default=True	Execution success flag
created_at	DateTime	Auto Now Add	Telemetry timestamp

Chapter 7: Artificial Intelligence & Machine Learning Subsystems

7.1 Groq Cloud LLM Architecture

LEX REGIS integrates with Groq's low-latency inference engine using model endpoints such as `openai/gpt-oss-120b`. The platform guarantees structural reliability by utilizing JSON-mode prompting and strict schema enforcement.

```
sequenceDiagram
    autonumber
    actor Citizen as Citizen / Litigant
    participant View as IntakeWizardView
    participant Log as AIRequestLog (Telemetry)
    participant GroqAPI as Groq Cloud Engine
    participant DB as System Database

    Citizen->>View: Submit Grievance (Description, Location, Urgency, Docs)
    View->>DB: Save LegalIssue Record (status='PENDING_ANALYSIS')
    View->>Log: Initialize AIRequestLog Record

    Note over View,GroqAPI: System Prompt enforces structured JSON output schema
    View->>GroqAPI: POST /chat/completions (System Prompt + Grievance Text)

    activate GroqAPI
    GroqAPI-->>View: Return Validated JSON Payload (Statutory Acts, Next Steps, Estimates)
    deactivate GroqAPI

    View->>Log: Update Latency, Token Usage, Status Code 200
    View->>DB: Create AIAalysis Record
    View->>DB: Update LegalIssue (status='ANALYZED')
    View-->>Citizen: Redirect to Analysis Dashboard with Next Steps & Lawyer Matches
```

7.2 System Prompts & Structured Extraction Schema

The system prompt for the AI Legal Triage engine is rigorously engineered:

```
{
  "category": "Broad legal domain (e.g., Civil, Criminal, Corporate, Family)",
  "practice_area": "Specific practice area (e.g., Property Dispute, Employment Law)",
  "subcategory": "Granular issue (e.g., Unlawful Tenant Eviction)",
  "complexity": "Low | Medium | High | Critical",
  "recommended_lawyer_type": "Designation of counsel required",
  "recommended_court": "Jurisdictional forum (e.g., District Civil Court)",
  "suggested_documents": ["List of evidentiary documents needed"],
  "possible_acts": ["Relevant statutory enactments (e.g., Transfer of Property Act 1882)"],
  "possible_sections": ["Specific sections (e.g., Section 106)"],
  "important_keywords": ["Extracted legal entities & terms"],
  "suggested_next_steps": ["Procedural recommendations for litigant"],
  "timeline_estimate": "Predicted duration (e.g., 6-12 Months)",
  "cost_estimate": "Estimated budget (e.g., ₹50,000 - ₹1,00,000)",
  "risk_level": 45,
  "confidence_score": 92
}
```

7.3 Machine Learning Predictive Modeling

The `apps.ml` module establishes the groundwork for continuous judicial machine learning:

1. **Case Duration Predictor (`CasePrediction.duration_days_predicted`)**: Regression pipeline taking case

complexity, court venue, number of parties, and historical practice area disposal times as features.

2. **Success & Risk Estimators** (`success_probability` , `risk_score`): Multi-factor scoring assessing statutory alignment, documentary evidence completeness, and prior judicial precedents.

3. **Court Workload Indexing** (`CourtWorkload.congestion_index`): Real-time queuing model evaluating pending dockets against disposal velocity across district court rooms.

Chapter 8: Cryptographic Document Vault & Blockchain Ledger Subsystem

8.1 Evidentiary Integrity & Cryptographic Checksums

To prevent spoliation of evidence, every uploaded file undergoes an automated checksum generation workflow:

$$\text{Checksum} = \text{SHA-256}(\text{Binary Content Stream})$$

The resulting 64-character hexadecimal digest is permanently attached to the `Document` model. If a document's binary stream is altered on the storage disk, any subsequent verification detects a hash mismatch immediately.

8.2 The Blockchain Verification Ledger (`BlockchainRecord`)

The blockchain subsystem simulates decentralized, tamper-proof notarization:

```
graph LR
    Doc[Uploaded Legal Document] --> HashGen[SHA-256 Engine]
    HashGen --> DocHash["Payload Hash: e.g. e3b0c44298fc1c149afbf4..."]
    DocHash --> LedgerService[Blockchain Verification Service]
    LedgerService --> PayloadJSON[Canonical JSON Stringification]
    PayloadJSON --> MerkleLeaf[Merkle Tree Leaf / Block Entry]
    MerkleLeaf --> TxGen[Simulate Ethereum Transaction 0x7f9a...]
    TxGen --> RecordDB[(BlockchainRecord Ledger)]
    RecordDB --> DocUpdate[Update Document with TxHash & Verified Timestamp]
```

- **Generic Foreign Key Binding:** Enables notarizing documents, case timeline events, hearing outcomes, or entire case snapshots.
- **Payload Non-Repudiation:** The exact JSON snapshot of the object attributes at the moment of verification is preserved alongside the transaction hash and block number.

Chapter 9: Real-Time Communication & Notification Architecture

9.1 ASGI & WebSocket Channel Pipeline

Standard HTTP request-response cycles are inadequate for live litigation interactions. LEX REGIS deploys an asynchronous pipeline:

```
graph TD
    BrowserClient[Web Browser Client] <-->|WebSocket wss://.../ws/chat/conversation_id/| DaphneServer[DaphneServer]
    DaphneServer <--> AuthMiddleware[Channels Auth & Session Middleware]
    AuthMiddleware <--> Consumer[ChatConsumer (AsyncWebSocketConsumer)]

    Consumer <--> RedisBus[Redis Channel Layer: chat_room_group]
    Consumer <--> SyncToAsync[database_sync_to_async Bridge]
    SyncToAsync <--> DjangoORM[Django ORM / SQLite / PostgreSQL]
```

9.2 Real-Time Event Dispatch

- 1. Connection Lifecycle:** When a user opens a case discussion, the client establishes a WebSocket connection. `ChatConsumer` validates whether the user is an assigned participant (`is_participant`).
- 2. Broadcast Protocol:** Upon receiving a message frame, `ChatConsumer` commits the `Message` instance to the database, serializes sender metadata, and broadcasts the frame across the Redis room group.
- 3. Instant Notification Trigger:** Simultaneously, the system dispatches an in-app `Notification` instance to alert offline participants upon their next login.

Chapter 10: Security, Authorization & Legal Regulatory Compliance

10.1 Role-Based Access Control (RBAC) & Object Permissions

Access control is implemented across multiple defensive layers:

- **View-Level Guards:** Class-based mixins (`LoginRequiredMixin`, `RoleRequiredMixin`).
- **REST API Permissions:** Custom DRF permission classes (`IsVerifiedLawyer`, `IsCitizen`, `IsAdministrator`, `IsOwner`).
- **Object-Level Isolation:** Selectors explicitly scope queries to the authenticated user (e.g., a citizen can only fetch cases where `client__user=request.user`, and a lawyer can only view cases where `assigned_lawyer__user=request.user`).

10.2 Statutory & Indian Regulatory Compliance

- **Identity & Legal Validations:** Custom regex validators for Indian identity documents (Aadhaar, PAN) and State Bar Council enrollment numbers.
- **Session Security:** Strict session settings (`SESSION_EXPIRE_AT_BROWSER_CLOSE = True`, `SESSION_COOKIE_AGE = 28800` / 8 hours, `SESSION_COOKIE_HTTPONLY = True`).

- **Data Protection:** Passwords secured with PBKDF2 with SHA-256 hashing. All sensitive files uploaded to structured, protected media subdirectories.

Chapter 11: REST API & WebSocket Protocol Specification

11.1 Key REST Endpoints

HTTP METHOD	ENDPOINT URI	DESCRIPTION	AUTH REQUIRED
POST	/api/v1/accounts/register/	Register new user account	Public
POST	/api/v1/accounts/token/	Obtain JWT access/refresh token pair	Public
GET	/api/v1/cases/	List all cases accessible to user	JWT / Session
POST	/api/v1/cases/	Create a new case docket	JWT / Session
GET	/api/v1/cases/{id}/	Retrieve full case details	JWT / Session
GET	/api/v1/documents/	List documents for case	JWT / Session
POST	/api/v1/documents/upload/	Multipart file upload with checksum	JWT / Session
GET	/api/v1/hearings/	Retrieve hearing docket	JWT / Session
POST	/api/v1/hearings/schedule/	Schedule a new court hearing	JWT / Session
POST	/api/v1/ai/chat/	Send message to Lex AI Assistant	JWT / Session

11.2 WebSocket Event Contract

Connection Endpoint:

```
ws://<host>/ws/chat/<conversation_id>/
```

Client Outgoing Message Frame:

```
{
  "message": "Please review the updated affidavit draft filed today."
}
```

Server Broadcast Message Frame:

```
{
  "message": "Please review the updated affidavit draft filed today.",
  "sender_id": "4b6f12a3-89c0-4e12-b5e1-8829f0123456",
  "sender_name": "Adv. Priya Sharma",
  "created_at": "11:45 AM"
}
```

Chapter 12: User Experience, Workflows & Interface Walkthrough

12.1 Citizen User Journey

1. **Public Landing (/)**: The citizen learns about the platform's services and accesses the AI Intake button.
2. **AI Intake Wizard (/intake/wizard/)**:
 - *Step 1*: Input legal issue description, incident date, and jurisdiction.
 - *Step 2*: Upload supporting evidence documents.
 - *Step 3*: View dynamic AI analysis (applicable Acts, timeline, complexity, estimated cost).
3. **Advocate Selection (/intake/recommendations/<issue_id>/)**: The citizen reviews recommended advocates with match percentages and fee structures.
4. **Consultation Booking**: Citizen clicks "Request Consultation". A `ConsultationRequest` and preliminary `Case` docket are automatically created.
5. **Case Tracking Workspace (/cases/<id>/)**: The citizen tracks hearings, views filed documents, monitors payment milestones, and chats with counsel.

12.2 Advocate User Journey

1. **Advocate Onboarding (/portal/lawyer/register/)**: 9-step registration capturing Bar Council enrollment, practice areas, office location, and fee schedules.
 2. **Lawyer Dashboard (/portal/lawyer/dashboard/)**:
 - View pending consultation requests from the intake marketplace.
 - Monitor today's court hearings, upcoming client meetings, and active tasks.
 3. **Matter Acceptance**: Clicking "Accept Matter" automatically transitions the case to `LAWYER_ASSIGNED` and opens a dedicated real-time chat room.
 4. **Case Management**: Upload pleadings, schedule court hearings, record adjournment reasons, and log case financials.
-

Chapter 13: Testing, Verification & Quality Assurance Report

13.1 Testing Strategy

The testing harness combines unit tests, service layer tests, permission guards, and API endpoint validations:

```
graph TD
  TestRunner[Django Test Runner / Pytest] --> UnitTests[Unit Tests: Models, Clean Methods, Validator]
  TestRunner --> ServiceTests[Service Tests: CaseCreationService, BlockchainVerificationService, AIC]
  TestRunner --> SelectorTests[Selector Tests: CaseSelector, UserSelector, DocumentSelector]
  TestRunner --> PermissionTests[Permission Tests: RBAC, Role Authorization, Object Ownership]
  TestRunner --> APITests[API Tests: DRF ViewSets, JWT Authentication, Serializer Validation]
```

13.2 Key Test Suites in Codebase

- `apps/accounts/tests/test_managers.py` : Verifies `UserManager.create_user` and `create_superuser` behaviors.
- `apps/accounts/tests/test_permissions.py` : Validates role-based access restrictions.
- `apps/cases/tests/test_services.py` : Tests atomic case creation, state transitions, and timeline generation.
- `apps/cases/tests/test_selectors.py` : Validates optimized querysets and search filters.
- `apps/documents/tests/test_services.py` : Asserts SHA-256 checksum generation and document versioning.
- `apps/hearings/tests/test_services.py` : Validates hearing scheduling, conflict detection, and adjournment counts.
- `apps/common/tests/test_validators.py` : Confirms regex enforcement for Aadhaar, PAN, and Bar Council numbers.

Chapter 14: Deployment, DevOps & Infrastructure Guide

14.1 Production Architecture Blueprint

```
graph TD
  Internet[Public Internet Traffic] --> Cloudflare[Cloudflare CDN & DDoS Protection]
  Cloudflare --> Nginx[Nginx Reverse Proxy & SSL Termination]

  Nginx -->|HTTP Requests / Static / Media| StaticMedia[Static & Media Storage]
  Nginx -->|Dynamic Web / API Traffic| DaphneCluster[Daphne ASGI Server Cluster]
  Nginx -->|WebSocket Traffic wss://| DaphneCluster

  DaphneCluster <--> RedisInstance[(Redis 7.x: Channels & Cache)]
  DaphneCluster <--> PostgresDB[(PostgreSQL 16 Enterprise Database)]

  DaphneCluster -.→|Async Tasks / Reminders| CeleryWorker[Celery Task Workers]
  CeleryWorker <--> RedisInstance

  DaphneCluster -->|LLM Inference Requests| GroqCloud[Groq Cloud LLM API]
```

14.2 Environment Configuration Checklist

Create a production `.env` file based on `.env.example` :

```

SECRET_KEY=production-crypto-strong-secret-key-32-chars-min
DEBUG=False
ALLOWED_HOSTS=lexregis.domain.com,api.lexregis.domain.com

# Database (PostgreSQL)
DB_ENGINE=django.db.backends.postgresql
DB_NAME=lex_regis_prod
DB_USER=lex_admin
DB_PASSWORD=secure_production_password
DB_HOST=127.0.0.1
DB_PORT=5432

# Redis Channel Layer
REDIS_URL=redis://127.0.0.1:6379/0

# Groq Cloud AI Engine
GROQ_API_KEY=gsk_your_groq_production_api_key
GROQ_MODEL=openai/gpt-oss-120b

# Email Subsystem
EMAIL_HOST=smtp.sendgrid.net
EMAIL_PORT=587
EMAIL_USE_TLS=True
EMAIL_USER=apikey
EMAIL_PASSWORD=your_sendgrid_key

```

14.3 Database Backup & Recovery Procedures

The platform includes automated maintenance scripts under `scripts/`:

- **Database Backup** (`scripts/backup_database.py`): Dumps database tables and media file archives with timestamped rotation.
- **Database Restoration** (`scripts/restore_database.py`): Rebuilds schemas and imports verified snapshots atomically.

Chapter 15: Conclusion, Limitations & Future Roadmap

15.1 Project Achievements

LEX REGIS successfully delivers a state-of-the-art, comprehensive LegalTech platform:

1. **Democratized Legal Access:** Enables citizens to articulate grievances and receive structured, actionable legal intelligence within seconds.
2. **End-to-End Practice Management:** Provides advocates and law firms with a modern, high-efficiency docket, case state machine, and client collaboration suite.
3. **Evidentiary Integrity:** Establishes non-repudiation and tamper evidence through SHA-256 cryptographic hashing and immutable ledger recording.
4. **Architectural Rigor:** Demonstrates enterprise software engineering standards via modular monolithic Django architecture, Service-Selector separation, and complete asynchronous WebSocket support.

15.2 Current Limitations

- **External Blockchain Mainnet Integration:** Currently uses an enterprise simulation of Ethereum/IPFS transactions rather than live public gas-fee mainnet submission.

- **OCR Engine Dependency:** Full-text PDF OCR requires server-side Tesseract-OCR binary installation in deployment environments.

15.3 Future Roadmap

- **Phase II (Q4 2026):** Native integration with the Indian e-Courts API for live CNR status fetching and automatic cause-list synchronizations.
- **Phase III (Q1 2027):** Smart Contract legal escrow system for client-advocate fee milestones.
- **Phase IV (Q2 2027):** Multilingual voice intake assistant supporting regional languages (Hindi, Tamil, Telugu, Marathi, Bengali).

Appendix: Directory Structure & File Index

```
lex_regis/
├── apps/
│   ├── accounts/           # Identity, Profiles, Security Settings, Auth Services
│   │   ├── models/         # User, Profile, ProfessionalProfile, Organization, Address
│   │   ├── services/       # Registration, Auth, Profile, Verification Services
│   │   ├── selectors/      # User and Profile Query Optimizers
│   │   └── presentation/   # Multi-step Registration, Profile Views, Security Views
│   ├── intake/            # AI Intake Wizard, LegalIssue, AIAAnalysis, Recommendations
│   ├── cases/             # Core Case Domain, State Machine, Financials, Parties, Timeline
│   ├── documents/         # Document Vault, Checksums, Versions, Master Classifications
│   ├── hearings/         # Hearing Schedules, Adjournments, Court Rooms, Sequence Counters
│   ├── ai/               # Groq LLM Client, Prompt Templates, Telemetry Logs, AI Chat
│   ├── ml/               # Case Predictions, Court Workload Statistics
│   ├── blockchain/       # Blockchain Records, Cryptographic Verification Service
│   ├── communication/    # Channels ASGI WebSockets, ChatConsumer, Conversation Rooms
│   ├── notifications/    # Multi-Channel In-App Alerts & Notification Services
│   ├── dashboard/        # Role-Specific Portals, Global Search, KPI Presenters
│   ├── analytics/        # National & State Judicial Metrics, Demo Data Provider
│   ├── lawyer_portal/    # 9-Step Advocate Onboarding & Lawyer Operations Workspace
│   ├── public_site/      # Marketing Landing Page, Public Directory, Legal FAQ
│   └── common/           # BaseModel, Mixins, Validators, HTMX Helpers, Custom Exceptions
├── config/               # Settings (base, development, production), ASGI, WSGI, URLs
├── templates/            # 45+ Production HTML Templates (Bootstrap 5, HTMX, Alpine.js)
├── static/               # Custom CSS, Glassmorphism Styling, JavaScript Modules
├── media/               # Vault file storage (Evidence, Profiles, Certificates)
└── scripts/             # Backup, Recovery, Superuser Setup & Data Seeders
```

End of Report — LEX REGIS Enterprise LegalTech Platform